Business Requirement Documentation

Created By:- Ashok Kumar Bhariya
Project Name:- MindForge AI - Personalized Learning App

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1. Executive Summary

1.1 Background

The rapid growth of online education has made learning resources more accessible than ever before. However, most existing learning platforms follow a standardized approach, offering the same content and learning paths to every learner regardless of their prior knowledge, learning pace, or career objectives. This often results in low engagement, inefficient learning, and poor knowledge retention.

MindForge AI was developed to address these challenges by providing an AI-powered personalized learning platform that adapts to individual learning goals, assesses existing knowledge, recommends customized learning paths, and continuously tracks learner progress. By integrating Artificial Intelligence (AI), Large Language Models (LLMs), and semantic search technologies, the platform delivers tailored educational experiences that improve both learning efficiency and long-term retention.

1.2 Business Drivers

The project is driven by the following business needs:

- Increasing demand for personalized learning experiences in the EdTech industry.
- Need to improve learner engagement and course completion rates.
- Reduce the time learners spend searching for relevant educational resources.
- Improve knowledge retention through intelligent revision and progress tracking.
- Automate the creation of learning roadmaps, quizzes, and study materials using AI.
- Provide actionable learning analytics to support continuous improvement.

1.3 Key Deliverables

The project will deliver the following core capabilities:

- Secure user authentication and profile management.
- AI-powered SkillScan for assessing learner knowledge.
- Personalized learning roadmap generation.
- AI-generated quizzes and study resources.
- Visual Concept Cards for effective revision.
- Knowledge Decay Tracker to monitor retention.
- Concept Confusion Detector using semantic similarity.
- Interactive learner dashboard with performance analytics.
- Exportable AI-generated study material in PDF format.

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2. Project Objectives

The primary objectives of the MindForge AI project are:

- Develop a personalized AI-driven learning platform that adapts content based on individual learner goals and knowledge levels.
- Improve learner engagement through interactive quizzes, personalized roadmaps, and AI-generated educational resources.
- Enhance long-term knowledge retention using revision recommendations and spaced learning techniques.
- Reduce manual effort in planning and managing learning journeys through automation.
- Provide learners with clear insights into their progress using data-driven analytics.
- Deliver a scalable web-based solution that can support future enhancements and additional learning domains.

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3. Project Scope

3.1 In Scope

The following functionalities are included in the scope of this project:

- User Registration and Login
- User Profile Management
- SkillScan AI Assessment
- Personalized Learning Roadmap Generation
- AI-Based Learning Resource Recommendations
- Quiz Generation and Evaluation
- Visual Concept Cards
- Knowledge Decay Tracking
- Concept Confusion Detection
- Progress Dashboard
- PDF Study Guide Generation

3.2 Out of Scope

The following features are not included in the current project release:

- Mobile application support
- Live tutoring or video conferencing
- Multi-language content generation
- Learning Management System (LMS) integrations
- Social learning communities and discussion forums
- Payment gateway integration
- Offline learning capabilities

3.3 Project Constraints

Constraint	**Description**
Timeline	Developed during a 12–13 day hackathon/internship.
Budget	Limited to available hackathon resources and infrastructure.
AI Dependency	Core functionality depends on third-party LLM services.

Platform	Initial release supports web browsers only.

4. Stakeholder Identification

Stakeholder	Role	Responsibilities	Expectations
Training Mentor / Project Guide	Project Sponsor	Defines project vision, reviews deliverables, and approves major requirements	Well-documented requirements and successful project completion
Business Analyst	Requirements Management	Gather, analyze, document, and validate business requirements	Clear, accurate, and traceable documentation
Development Team	Solution Development	Implement application features, APIs, and business logic	Complete and implementable functional requirements
UI/UX Designer	User Interface Design	Design intuitive user interfaces and improve user experience	Clear screen requirements and user-friendly designs
QA Tester	Quality Assurance	Validate implemented features against documented requirements	Stable and defect-free application
End Users (Students/Learners)	Primary Users	Use the platform for personalized learning	Personalized learning experience and accurate recommendations

5. Business Requirements

Requirement ID	Business Requirement	Priority
BR-001	The system shall allow users to register and securely authenticate.	High
BR-002	The system shall assess the learner's current knowledge using SkillScan AI.	High
BR-003	The system shall generate personalized learning roadmaps based on learner goals and assessment results.	High
BR-004	The system shall recommend AI-generated learning resources relevant to the learner's roadmap.	High
BR-005	The platform shall generate quizzes to evaluate learner understanding after completing learning modules.	High
BR-006	The system shall generate Visual Concept Cards for quick revision.	Medium
BR-007	The system shall monitor knowledge retention using a Knowledge Decay Tracker and recommend revision sessions.	High
BR-008	The system shall identify concept confusion through semantic similarity analysis and provide targeted recommendations.	Medium

BR-009	The platform shall provide an interactive dashboard displaying learner progress, achievements, and analytics.	High
BR-010	The system shall allow learners to export AI-generated study materials as PDF documents.	Medium

5.1 Supporting Wireframes

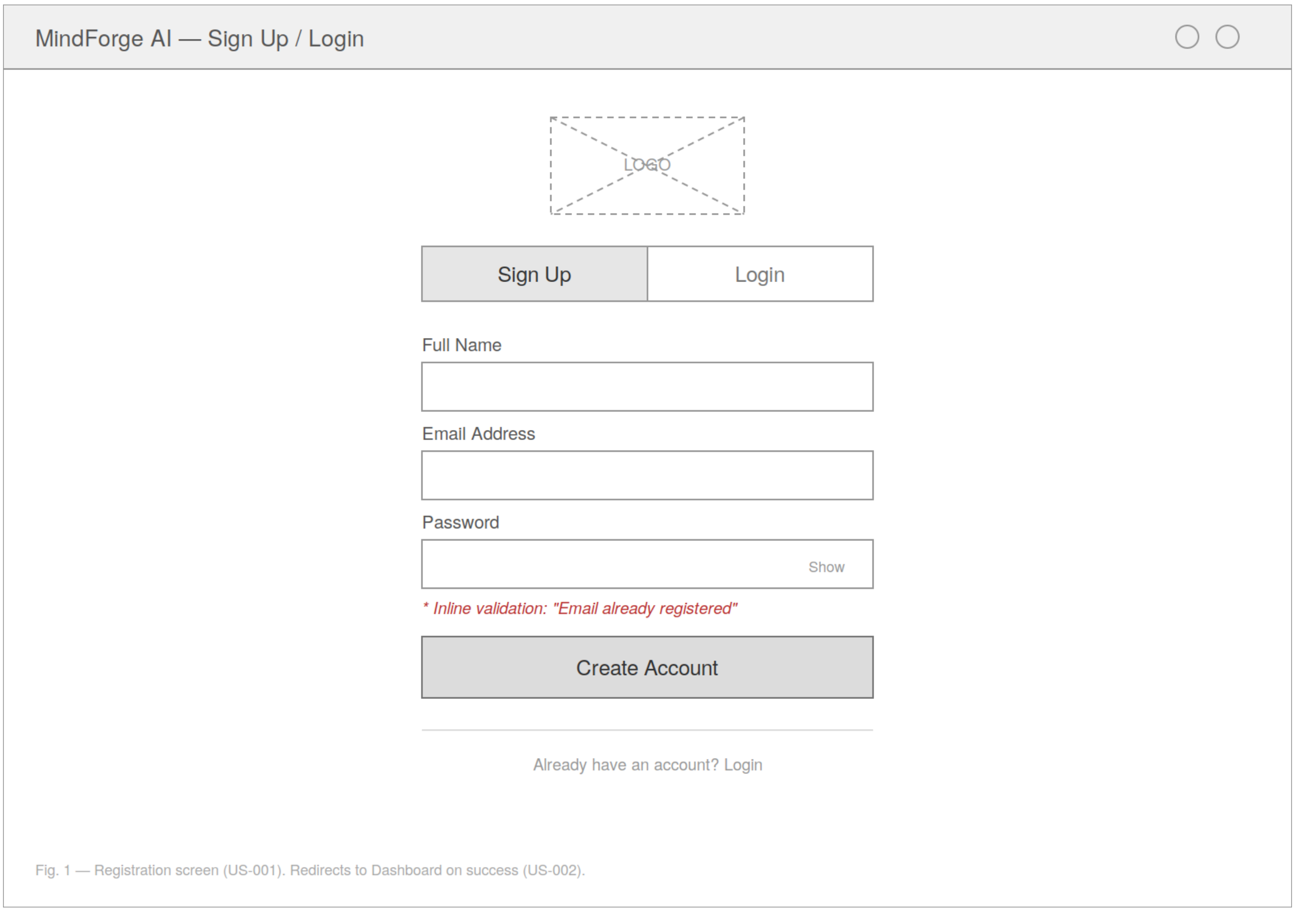

Fig. 1. Wireframe for BR-001

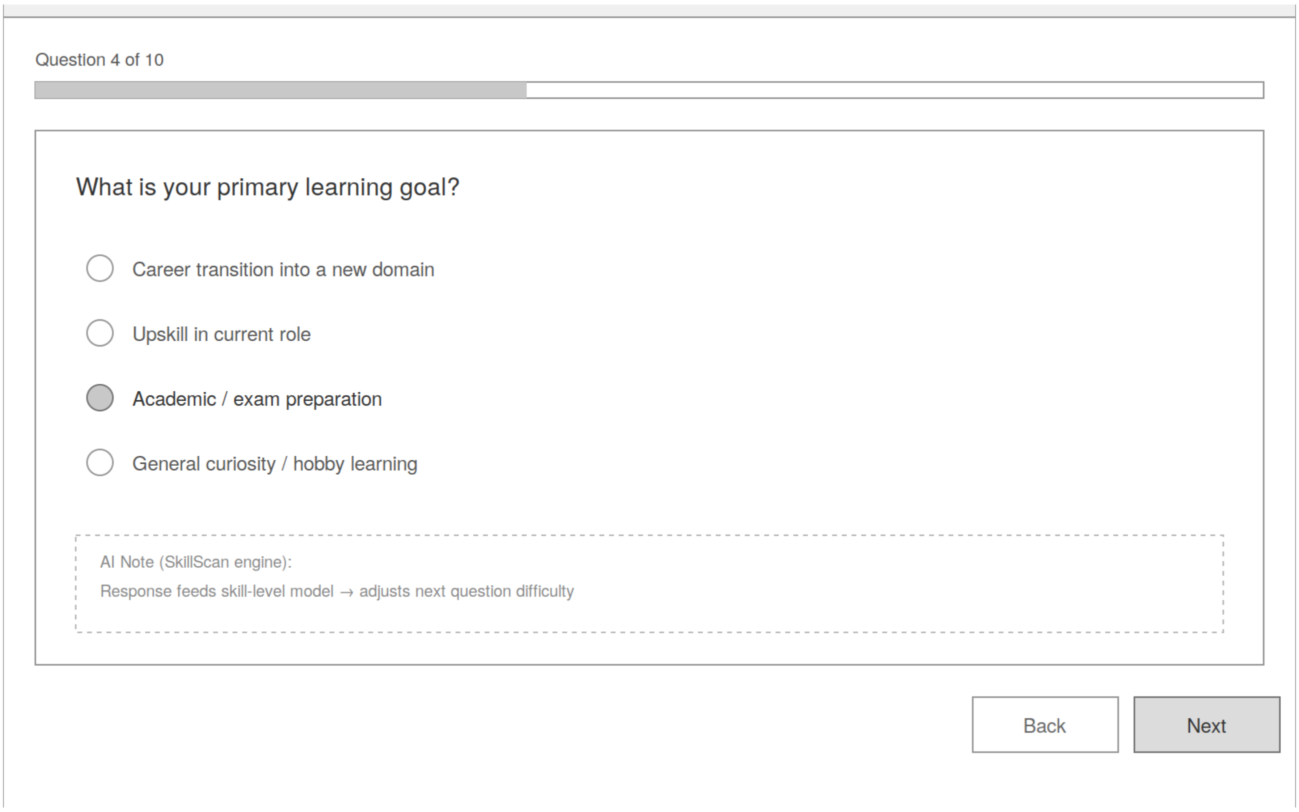

Fig. 2. Wireframe for BR-002

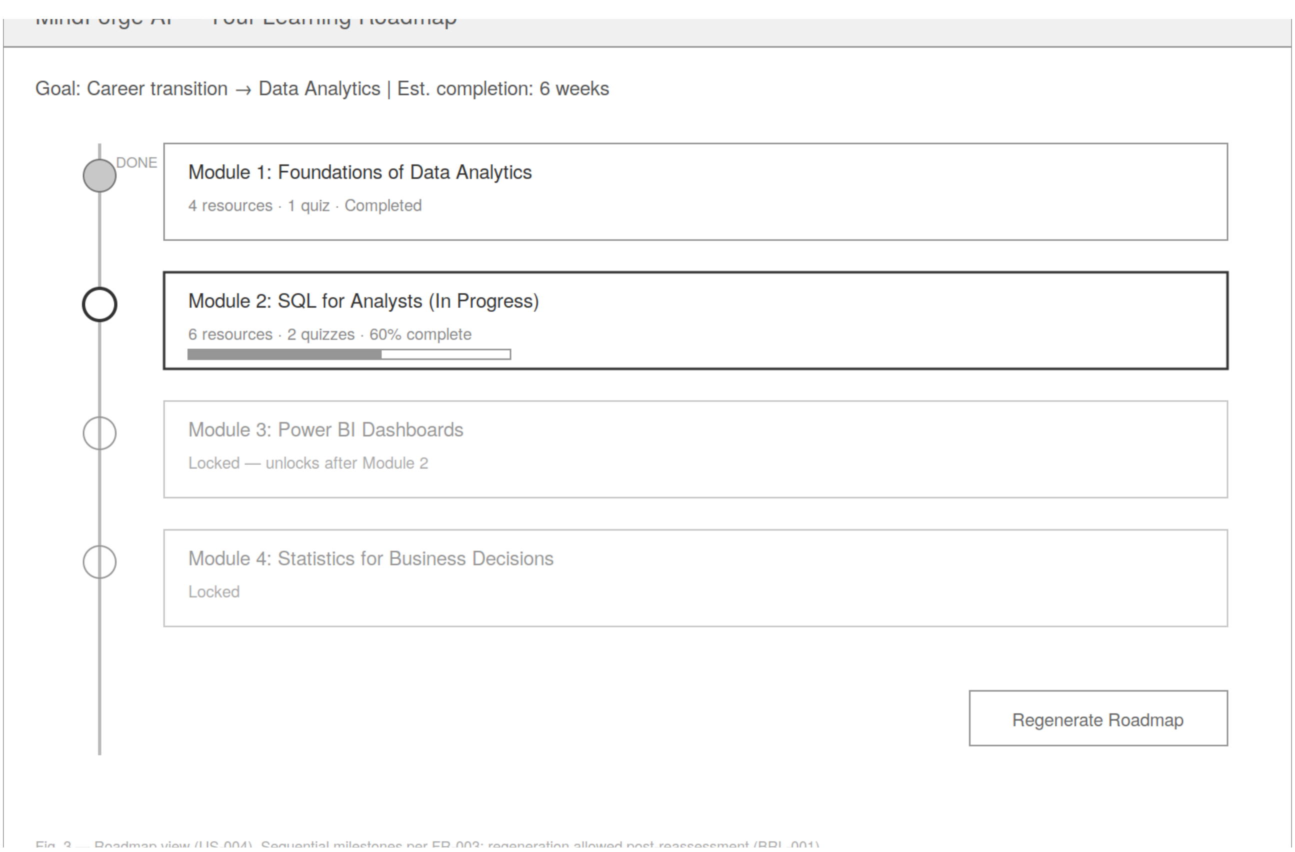

Fig. 3. Wireframe for BR-003

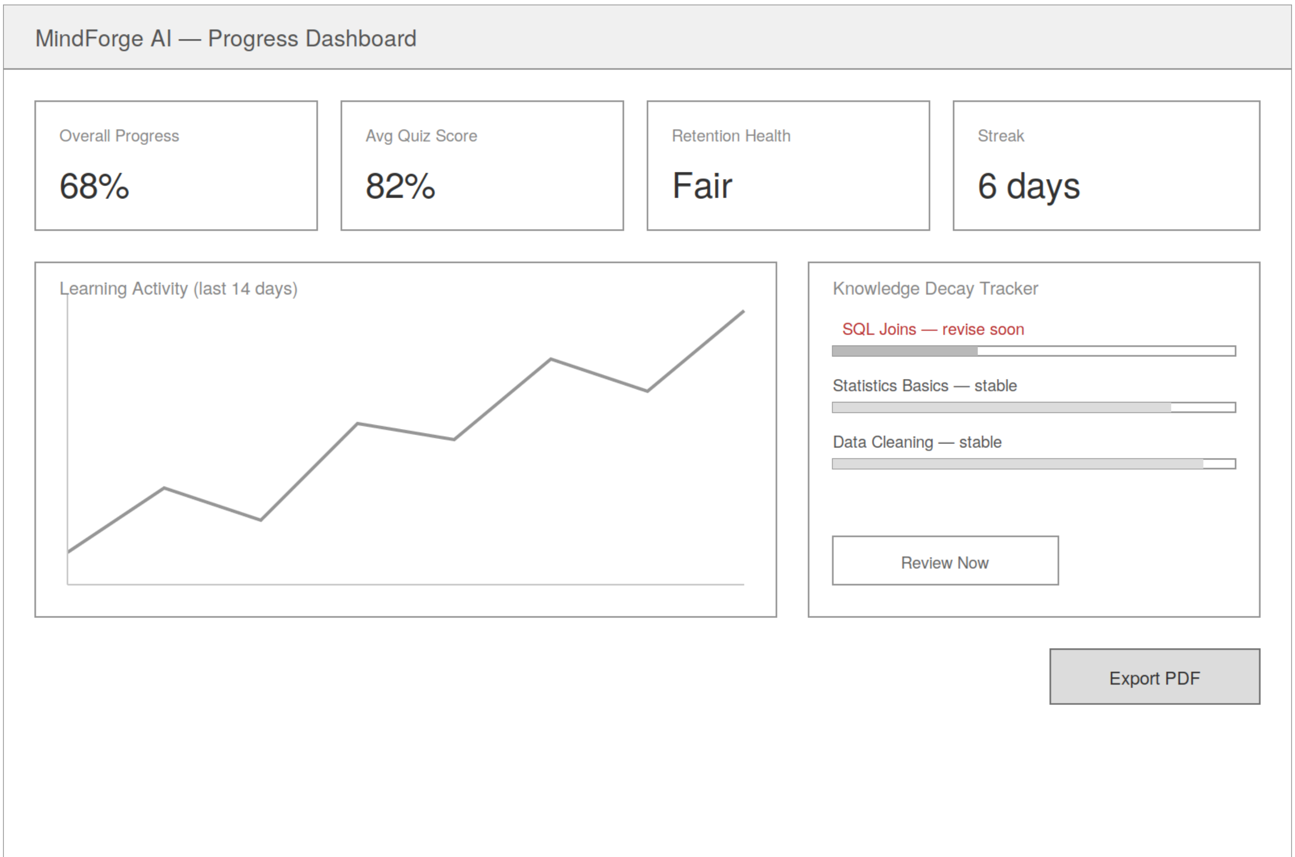

Fig. 4 — Dashboard (US-009). Aggregates FR-008 metrics; decay panel reflects FR-006 output.

Fig. 4. Wireframe for BR-009

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6. Assumptions & Constraints

Assumptions

- Users have a stable internet connection.
- AI services and APIs are available throughout platform usage.
- Users provide accurate learning goals and preferences.
- Educational resources required for recommendation remain accessible.

Constraints

- Limited project duration and development resources.
- Dependence on third-party AI APIs.
- Initial release supports English-language content only.
- Platform is designed for desktop and modern web browsers.

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7. Risk Management

Risk	Probability	Impact	Mitigation Strategy
AI service downtime	Medium	High	Implement retry logic and fallback messaging
AI-generated content inaccuracies	Medium	Medium	Enable user feedback and manual verification
Performance degradation with increased users	Low	High	Optimize database queries and implement caching
Delays due to hackathon timeline	Medium	Medium	Prioritize MVP features and iterative delivery0

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8. Acceptance Criteria

The project will be considered successfully delivered when:

- All High-priority business requirements are implemented and validated.
- Users can register, authenticate, and manage their profiles successfully.
- SkillScan AI generates meaningful learner assessments.
- Personalized learning roadmaps are created based on user inputs.
- AI-generated quizzes, learning resources, and Visual Concept Cards function as intended.
- Progress dashboard accurately reflects learner performance and activity.
- Knowledge Decay Tracker provides timely revision recommendations.
- PDF generation is operational.
- Stakeholders review and approve the delivered solution.

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9. Appendices

Appendix A – Glossary

Term	Definition
AI	Artificial Intelligence
LLM	Large Language Model
BRD	Business Requirements Document
API	Application Programming Interface
FastAPI	Python framework used for backend development
PostgreSQL	Relational database used for structured data
Qdrant	Vector database used for semantic search
SkillScan AI	AI-powered learner assessment module